Chamith Dilshan Ranathunga

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Research interests : Prosody-Aware Speech Synthesis | Bio-Signal Processing | Applied Machine Learning in Healthcare

Education

- **University of Melbourne** — Australia
  Ph.D. Researcher — *May 2026 - Present*
  - **Lab:** Neurobionics Lab
  - **Research Focus:** Prosody-aware speech synthesis from ECoG
  - **Advisor:** Prof. Sam John
- **University of Moratuwa** — Sri Lanka
  B.Sc. Engineering Honours (Biomedical Engineering) | First Class (CGPA: 3.84/4.2) — *2020 - 2024*

Experience

- **Lead AI Engineer** — Remote (Contract: UK)
  Braingaze (Spain) — *Sep 2024 - May 2026*
  - Engineered a multimodal ADHD diagnostic classifier in PyTorch using time-series data from a Posner cue task, training a Multiple Instance Learning (MIL) model on pupil dilation alongside Naive Bayes and Logistic Regression for eye synchrony.
  - Extracted robust diagnostic markers by fusing the independent models in logit space as normal distributions and validating against a real-world clinical patient dataset; resulted in a first-author publication in *Scientific Reports* in collaboration with Prof. Hans Supèr.
  - Directed clinical AI research pipelines, directly mentoring a total of 9 engineering interns and 1 software engineer throughout my tenure, including guiding the foundational development of an experimental Autism diagnostic classifier.
- **Software Engineer - Research and Development** — Sri Lanka
  Wavenet Pte. Ltd. — *Jun 2024 - Sep 2024*
  - Architected a Retrieval-Augmented Generation (RAG) agent using LlamaIndex for a major Tier-1 telecom client, vectorizing enterprise databases to provide call center agents with low-latency, real-time troubleshooting data.
  - Engineered time-series fraud detection models (ARIMA and LSTM) to identify Wangiri telecom fraud patterns, enhancing network security and mitigating revenue leakage.
  - Optimized speech-to-text pipelines by developing a voice activity detection system to isolate silent audio segments in call recordings, significantly reducing downstream transcription API costs.
- **Visiting Research Fellow (Internship)** — Singapore
  Singapore University of Technology and Design — *Jan 2023 - Jun 2023*
  - Conducted applied machine vision research under the supervision of Prof. Chau Yuen, focusing on real-time inference and edge computing architectures.
  - Engineered a real-time people tracking system by applying transfer learning to a Single Shot MultiBox Detector (SSD) for overhead camera feeds, achieving accuracy that resulted in a publication at IEEE Tencon 2024.
  - Developed an edge-optimized pipeline for illegal activity recognition from CCTV feeds using SSD, converting and deploying the model via TensorFlow Lite (TFLite) for real-time inference on an Intel NUC.
  - Built an unauthorized access detection system utilizing FaceNet to extract facial embeddings and accurately identify unauthorized personnel attempting to enter secured environments.
- **Visiting Instructor** — Sri Lanka
  University of Moratuwa — *Aug 2023 - Dec 2023*
  - Facilitated hands-on laboratory sessions for the EN2091 Laboratory Practice II module, instructing undergraduate students in applied electronics and signal processing experiments.
  - Mentored students through complex technical troubleshooting, bridging theoretical concepts with practical execution and evaluating experimental outcomes.

Publications

- **Real-world clinical validation of brainstem-based ocular biomarkers for ADHD classification** — *Jul 2026*
  - **Chamith Ranathunga**, August Romeo, Elizabeth Kilbey, Hans Supèr. "Real-world clinical validation of brainstem-based ocular biomarkers for ADHD classification in children and adults." *Scientific Reports* (2026).
- **Real-Time AI-Driven People Tracking and Counting Using Overhead Cameras** — *Dec 2024*
  - Ishrath Ahamed, **Chamith Dilshan Ranathunga**, Dinuka Sandun Udayantha, Benny Kai Kiat Ng, Chau Yuen. "Real-Time AI-Driven People Tracking and Counting Using Overhead Cameras." *IEEE Tencon Conference* (2024).

Projects

- **Ear-EEG hardware and interpretable deep learning in p300s based event classification** *Sep 2023 - April 2024*
  - Developed firmware for the ADS1299 analog-to-digital converter to accurately acquire high-fidelity EEG bio-signals from the ear.
  - Validated custom PCB designs and manufactured the wearable earpiece enclosure using 3D printing techniques.
  - Project awarded First Place by the IEEE Signal Processing Society for Best Undergraduate Thesis in Signal Processing.
- **Technology to enhance agriculture production - AgriMAC** *Apr 2021 - July 2022*
  - Developed an IoT climate replication system designed to recreate remote weather patterns inside closed agricultural chambers as a technological solution to food insecurity.
  - Programmed the embedded sensor arrays to capture real-time climate telemetry and engineered the control logic for the environmental actuators.
- **EMG signal acquisition device** *July 2023 - Nov 2023*
  - Designed and built a portable EMG bio-signal acquisition device strictly optimized to operate on a low-power, single 3.3V battery supply.
- **Venato - GPS tracker with SOS function** *Aug 2022 - Jan 2023*
  - Designed and manufactured a custom 3D-printed enclosure for a portable GPS and cellular telematics tracker, ensuring physical durability and field readiness for the hardware.

Honors and Awards

- **Undergraduate Thesis Project Competition 2024**
  First Place *2024*
  Awarded Best Undergraduate Thesis Project in the Signal Processing domain by the IEEE Signal Processing Society Chapter, Sri Lanka, for Ear-EEG biosignal hardware and interpretable deep learning integration.
- **Spark Challenge 2021/22**
  Finalist *2022*
  Selected as a finalist in a year-long innovation challenge organized by the E-Club, University of Moratuwa, developing SDG-aligned technology solutions.
- **Most Outstanding Continuous Project**
  First Place *2020*
  Awarded First Place at A2Con'20 (Leo District 306 A2) for "Sinhalen Python," an educational initiative teaching Python programming in native Sinhala.

Technical Skills

- **Languages:** Python, C, MATLAB
- **ML & Signal Processing:** PyTorch, MNE, TensorFlow Lite, OpenCV, Scikit-learn, SciPy, Pandas, NumPy
- **Hardware & Design:** SolidWorks, 3D Printing, PCB Validation & Debugging, Git

Community and Leadership

- **Council Member** *Aug 2023 - Aug 2024*
  IEEE Engineering in Medicine and Biology Society Student Branch Chapter *University of Moratuwa*
  - **Webmaster** (*Aug 2022 - Aug 2023*)
  - **Assistant Webmaster** (*Aug 2021 - Aug 2022*)
- **Assistant Pillar Head** *Mar 2021 - Apr 2022*
  Creative Design Pillar, Moraspirit Initiative *Sri Lanka*
- **Founder** *Mar 2019 - Present*
  Spectrum Show Foundation